

The image features the text "DL for ENG" in a bold, 3D, gold-colored font. The letters are floating on a vast, blue ocean with visible ripples and small waves. In the background, there are rolling blue hills or mountains under a sky filled with soft, grey clouds. The overall scene has a serene and expansive feel.

DL for ENG

LECTURE 8.2

Transient Heat

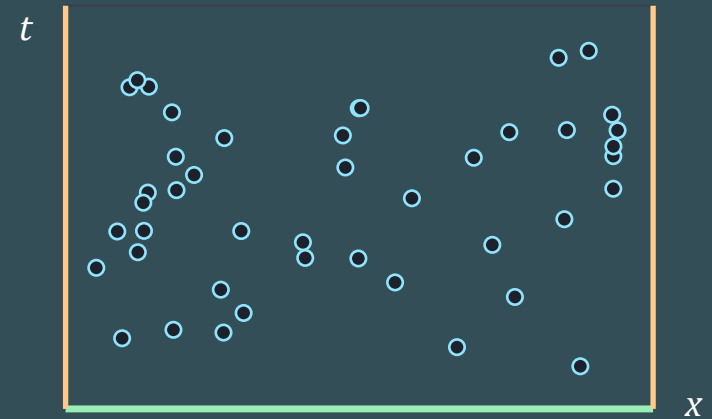
Remus Teodorescu

*With support from Research Assistant Noman Khan -
nomank@energy.aau.dk*

Aalborg University · ret@et.aau.dk

Deep Learning for Engineering · 2026

TIME · CAUSALITY · THE INVERSE PROBLEM



green: the field at $t = 0$ · amber: the walls

Time is one more input, and the whole block of space and time is trained at once. That buys a solution readable at any instant, and it costs the arrow of time: the loss does not know that early instants come first. The same plate then gives up its diffusivity to a few sensors.

Learning Objectives

SECTIONS, IN ORDER

- 1 **Storage and diffusion** — heat stored, how fast it spreads, the lumped model
- 2 **Time** — the transient equation, the space–time block, the initial field
- 3 **Causality** — why the loss does not know the arrow of time, and four fixes
- 4 **Plate and inverse** — the plate in time, checking it, and recovering α

READ, AND WHAT EACH ANSWERS

Incropera ch. 5	how does a body respond in time?
Liu ch. 4.1 – 4.3	what does time add to the plate?
Liu ch. 4.4	why are the early times fitted last?
Liu ch. 4.6	how is a transient checked, inverted?

Heat Storage

Heat storage — a material that warms up takes in energy: the heat stored in each volume is ρc_p times its temperature rise. In steady state nothing is stored. In a transient, the heat that arrives and is not conducted away is stored, and the temperature changes.

ENERGY STORED PER VOLUME

$$e = \rho c_p (T - T_0)$$

e the energy stored in J/m³, ρ the density, c_p the specific heat in J/(kg K)

- **steady** — nothing is stored: heat in equals heat out
- **transient** — the difference is stored, and T changes
- **examples** — a chip in a power pulse, a motor at start-up

STORAGE IN PYTORCH

```
e = rho * c_p * (T - T0)          # energy stored per volume, J/m^3
```

HEAT CAPACITY PER VOLUME

material	ρc_p , MJ/(m ³ K)
copper	3.45
aluminium	2.43
silicon	1.64
stainless steel	4.00
glass	2.10
water	4.18
air	0.0012

Typical values at room temperature. Solids and water store about the same per volume, two to four MJ/(m³ K); air stores three thousand times less.

Thermal Diffusivity

Thermal diffusivity — how fast a temperature change spreads through a material: the conductivity divided by the heat stored per volume, $\alpha = k/(\rho c_p)$. In a time t a change at the surface reaches a depth of about the square root of αt .

CONDUCT AGAINST STORE

$$\alpha = \frac{k}{\rho c_p}, \quad \delta \approx \sqrt{\alpha t}$$

α the diffusivity in m^2/s , δ the depth a change reaches in the time t

- **large α** — conducts well, stores little: fast
- **small α** — slow to respond: glass, water
- **example** — silicon: 0.30 mm in 1 ms, 9.57 mm in 1 s

DIFFUSIVITY IN PYTORCH

```
alpha = k / (rho * c_p)      # the diffusivity, m^2/s
delta = (alpha * t) ** 0.5    # the depth a change reaches in time t
```

DIFFUSIVITY OF COMMON MATERIALS

material	α , mm^2/s
copper	116
aluminium	98
silicon	91
stainless steel	3.75
glass	0.48
water	0.15
air	22

Typical values: the conductivities of the last lecture over the heat capacities of the last slide. Air conducts badly but stores almost nothing, so a change spreads through it faster than through steel.

Lumped Model

Lumped model — when the Biot number is small the body is nearly uniform in temperature, and one number describes it. The heat it stores changes at the rate its surface loses heat: an ordinary differential equation, whose solution decays exponentially with the time constant τ .

ONE TEMPERATURE FOR THE WHOLE BODY

$$\rho c_p V \frac{dT}{dt} = -h A (T - T_\infty), \quad \tau = \frac{\rho c_p V}{h A}$$

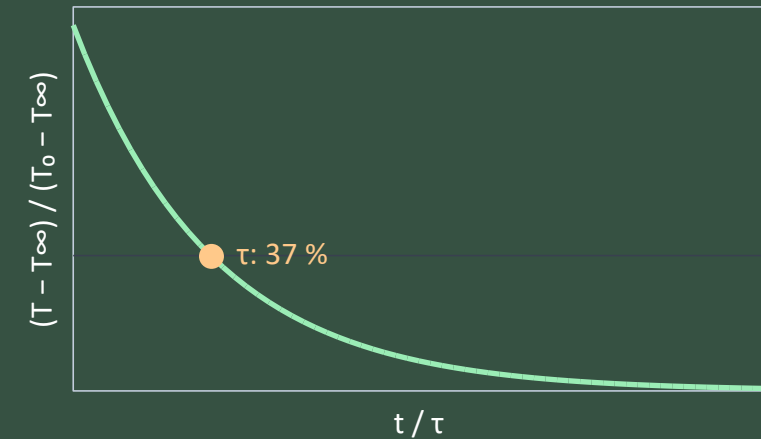
V the volume, A the cooled surface, h the heat transfer coefficient, T_∞ the fluid

- **valid** — $Bi \ll 1$: conduction inside is fast
- **answer** — an exponential: 37 % is left after τ
- **examples** — a die in still air, a thermocouple bead

THE LUMPED MODEL IN PYTORCH

```
tau = rho * c_p * V / (h * A)          # the time constant, s
T = T_inf + (T0 - T_inf) * torch.exp(-t / tau)  # the whole answer
```

A BODY COOLING IN A FLUID



1 mm of silicon in still air, $h = 25 \text{ W}/(\text{m}^2 \text{ K})$: $\tau = 65.60 \text{ s}$, while heat crosses the die in 10.93 ms. Their ratio is the die's Biot number, 1.67×10^{-4} .

Transient Heat in L8

Transient heat in L8 — this lecture solves the heat equation for a field in space and time: storage on one side, conduction on the other, the diffusivity as its one parameter. The lumped model has no field; it is used to estimate the time window, and no network is needed for it.

WHERE EACH IDEA ENTERS

	is	in L8.2
storage	the term $\partial T/\partial t$	new in the residual
diffusivity	the one parameter α	given, then found from sensors
lumped model	the limit $Bi \ll 1$	a first estimate of τ

WHAT IS LEFT OUT

- **melting, boiling** — latent heat and a moving front
- **properties that change with T** — a nonlinear equation
- **convection, radiation** — still boundary conditions

TWO TIME SCALES, ONE RATIO

	time
heat crosses the body	L^2/α
the surface empties it	$\rho c_p L/h$
their ratio	$Bi = hL/k$

Bi small: the inside keeps up with the surface, the field is flat, and the lumped model is enough. Bi not small: the field matters. That is the case this lecture solves, with the heat equation and a network.

Transient Conduction

Transient conduction — the heat equation with units:

$\partial T / \partial t = \alpha \nabla^2 T + Q / (\rho c_p)$, where $\alpha = k / \rho c_p$ is the diffusivity. It is parabolic, so besides the walls it needs the field at $t = 0$, and its time constant $\tau = L^2 / (d \pi^2 \alpha)$ says how long the transient lasts.

ONE NEW TERM, ONE NEW CONSTANT

$$\frac{\partial T}{\partial t} = \alpha (T_{xx} + T_{yy}) + \frac{Q}{\rho c_p} \quad \alpha = \frac{k}{\rho c_p} \text{ [m}^2/\text{s]}$$

$$\tau = \frac{L^2}{d \pi^2 \alpha}$$

α the diffusivity, ρc_p the heat capacity per volume, L the size, d the dimensions

THE TRANSIENT IN PYTORCH

```
xyt = torch.cat([x, y, t], 1)          # three inputs now
T = net(xyt)                            # the field in space and time
F = grad(T, t) - alpha * lap(T, xy) - Q_c # the residual; grad = autograd
tau = L**2 / (2 * pi**2 * alpha)        # the square's time constant
```

THE BENCHMARK MODE DECAYS



Ex_08.2's benchmark: one mode, decaying and keeping its shape. Amber is τ , orange 4τ , where it has settled. At $t = 1$ the amplitude is 3×10^{-9} .

Space-Time Sampling

Space-time sampling — the interior points carry a third coordinate, t , and the whole block $\Omega \times (0, t_{\text{end}}]$ is trained at once. The solution is continuous in time and can be read at any instant, and nothing in the loss requires early times to be right before late ones.

A BLOCK, NOT A MARCH

$(x, y, t) \in \Omega \times (0, t_{\text{end}}]$ $T(x, y, 0) = f_{IC}(x, y)$
 Ω the plate, t_{end} the end of the window, f_{IC} the initial field

- **no step size** — and no stability limit
- **any instant** — read the model, do not re-run it

THE BLOCK IN PYTORCH

```
xyt = torch.rand(4000, 3)          # x, y and t together
xyt[:, 2] *= t_end                  # the time window
xy0 = torch.cat([torch.rand(400, 2), zeros], 1) # the slice at t = 0
```

$\Omega \times \text{TIME}$, SAMPLED AS ONE BLOCK



the slice one condition, few points

the block the residual, many points

Cyan is the residual, green the initial field: it is outvoted.

Initial Condition

Initial condition — the field at $t = 0$. As a loss term it is one slice among a block of interior points and is easily outvoted; built into the trial solution as $(1 - t) f_{IC} + t(\dots)\mathcal{N}$ it holds exactly, which assumes a time axis scaled to zero to one.

A TERM, OR A FACTOR

$$T = (1 - t) f_{IC}(x, y) + t x(1 - x) y(1 - y) \mathcal{N}_{\hat{w}}$$

f_{IC} the initial field, $\mathcal{N}_{\hat{w}}$ the network; at $t = 0$ the second term vanishes

- **soft** — three terms compete, and the slice loses
- **hard** — two terms disappear, and $t = 0$ is exact

TALK · 2 MIN How would you explain why the initial condition is the term a network loses most often, in your own words? Then, in the card on the right, what does the $(1 - t)$ factor assume, and what would break if t_{end} were ten?

THE TRIAL SOLUTION, TERM BY TERM

$(1 - t) f_{IC}$ the initial field, fading

$t \cdot x(1-x) y(1-y)$ zero at $t = 0$ and on the walls

$\mathcal{N}(x, y, t)$ free everywhere else

Each factor switches off exactly one condition.

THE START IN PYTORCH

```
L_ic = (net(xy0) - f_ic).pow(2).mean() # soft: a term
T = (1-t)*f_ic + t*Dxy*net(xyt)      # hard: a factor
```

Causality

Causality — a solution at one instant follows from the earlier ones, but the loss sums over the block and treats $t = 0.01$ exactly as $t = 0.99$. A network can fit the late times first and carry an error at $t = 0$, the one instant that was known before training began.

THE ARROW OF TIME IS NOT IN THE LOSS

- **the loss** — one mean over the whole block
- **the symptom** — the largest error sits at $t = 0$
- **worse in reverse** — late times pull the early ones

TALK · 2 MIN How would you explain what causality means for a solution in time, in your own words? Then, in the plot on the right, why does the soft-start error grow towards $t = 0$, and would more interior points fix it?

SCHEMATIC · ERROR AGAINST TIME



Schematic: with a soft start the largest error sits at $t = 0$, the one instant that was known exactly before training began.

Causal Training

Causal training — four ways to put the arrow of time back into training: build the initial condition in, weight each instant by the residual accumulated before it, split the window and train the pieces in sequence, or sample the early times more densely.

WEIGHT AN INSTANT BY WHAT CAME BEFORE IT

$$w(t) = \exp\left(-\epsilon \int_0^t \mathcal{L}_{PDE}(s) ds\right)$$

$w(t)$ the weight of the instant t , ϵ how strict it is, $\mathcal{L}_{PDE}(s)$ the earlier residual

- ϵ — one more hyperparameter to set
- **the first choice** — hard-enforce, and tune nothing

CAUSAL WEIGHTS IN PYTORCH

```
res = residual(net, xyt).pow(2)           # per point, not averaged
acc = cumulative_by_time(res, xyt[:, 2])  # the residual before each t
loss = (torch.exp(-eps * acc).detach() * res).mean() # weight, then average
```

FOUR REMEDIES, IN ORDER

hard start removes the term; nothing to tune

causal weights effective; adds ϵ

time windows for long transients; more code

early points cheap; a partial fix

The order is the order to try them in. The first costs nothing, and on the problems of this block it is usually enough.

The Plate in Time

Transient plate — the plate of L8.1 with a time coordinate: uniform generation, the hole held at $T = 0$, insulated edges and a cold start. The trial solution $t\phi\mathcal{N}$ holds the hole and the initial field at once, and the edges stay a loss term.

ONE FACTOR, TWO CONDITIONS

$$T = t \phi(x, y) \mathcal{N}_{\hat{w}}(x, y, t)$$

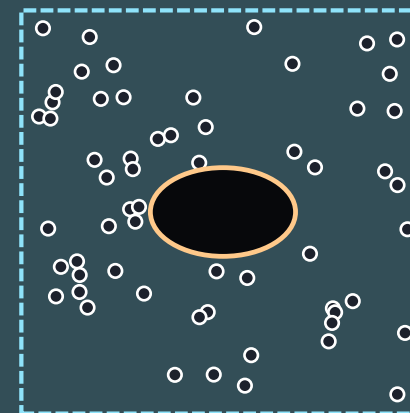
t holds the cold start, ϕ the level set holds the hole, $\mathcal{N}_{\hat{w}}$ the network

- **built in** — the hole, and the start at zero
- **a loss term** — the insulated edges, as before

THE PLATE IN PYTORCH

```
T = t * hole_multiplier(xy) * net(xyt)    # hole and cold start, exact
F = grad(T, t) - alpha * lap(T, xy) - Q_c  # the transient residual
loss = F.pow(2).mean() + w_N * L_edges     # the edges stay a term
```

THE PLATE, NOW WITH A CLOCK



starts	cold, at $T = 0$ everywhere
loses heat	through the hole only
settles in	0.13: only the hole cools it

Verification in Time

Verification in time — a transient has no single error. The relative error at late times can be large while the absolute error is tiny, so both are reported. The steady energy balance holds only once the field is steady, and a time history at the hottest point catches a wrong start.

TWO ERRORS, AND TWO CHECKS WITHOUT AN ANSWER

$$\varepsilon(t) = \frac{\|T_{\hat{w}}(\cdot, t) - T_{\text{exact}}(\cdot, t)\|_2}{\|T_{\text{exact}}(\cdot, t)\|_2}$$

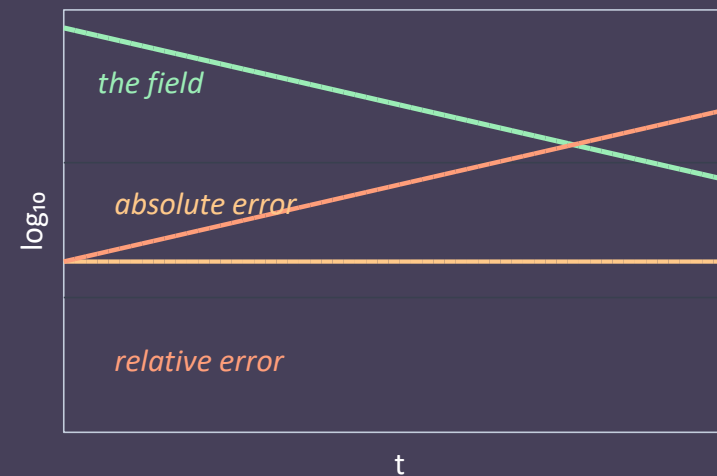
$T_{\hat{w}}$ the network, T_{exact} the reference, both at the instant t

- **report both** — relative and absolute, against t
- **without an answer** — the energy balance, and a history

ERROR AGAINST TIME IN PYTORCH

```
for t in times:
    # one instant at a time
    d = model_slice(t) - exact_slice(t) # the difference on a grid
    log(t, d.norm(), d.norm() / exact_slice(t).norm()) # absolute, relative
```

SCHEMATIC · WHY BOTH NUMBERS



The field decays, the absolute error does not, so the relative error climbs. Neither number alone says whether the answer is good.



Inverse Diffusivity

Inverse diffusivity — the diffusivity α made trainable beside the weights and identified from a measured cooling curve. It is optimised as $\log \alpha$ so that it cannot go negative, and only the early transient constrains it: once the field has decayed the data says nothing.

ONE RUN, TWO UNKNOWN S

$$\alpha = e^{\beta}, \quad \mathcal{L} = \mathcal{L}_{PDE} + \mathcal{L}_{data}$$

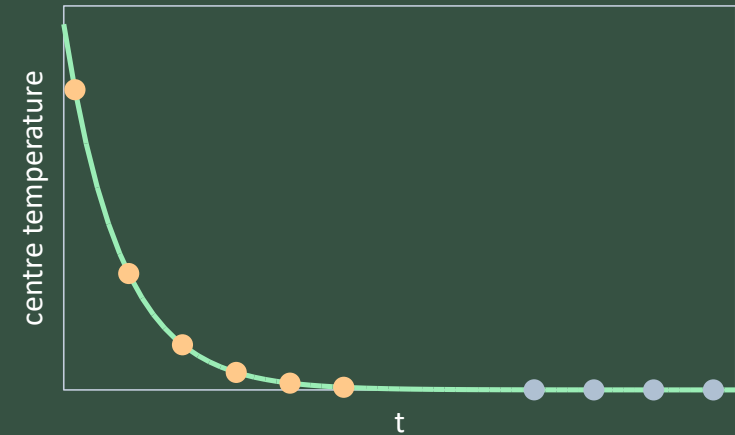
β the trained parameter, $\alpha = e^{\beta}$ the diffusivity, positive for any β

- **the data term** — what identifies α at all
- **the window** — early, where the curve still bends

THE INVERSE IN PYTORCH

```
beta = nn.Parameter(torch.zeros(1))      # log α, so α stays positive
opt = torch.optim.Adam([*net.parameters(), beta]) # trained together
loss = F(net, beta.exp()).pow(2).mean() + data(T, x_s) # residual + data
```

WHERE THE DATA CARRIES α



amber, early

the curve bends: α is identified

grey, late

the curve is flat: α is free

12 sensors, $t = 0.01$ to 0.25 , noise 0.002 .

Advanced Topics

RELATED TOPICS LEFT OUT, AND WHERE THEY MATTER

topic	what it adds	read	where it makes a difference
stiffness	scales that differ by decades	Liu ch. 4.10	thin films, fast pulses
other activations	sine and swish on a transient	Liu ch. 4.7.4	oscillating sources
convergence study	error against points, in x and t	Liu ch. 4.9.3	a design sign-off
walls that change	a flux that varies with time	Liu ch. 4.6.5	a switched source

WHY THEY ARE LEFT OUT

The plate of Ex_08.2 has one timescale, a constant source and a smooth start, so none of them is needed. Each returns when a problem brings it: the switched source in a power device, the separated scales in the flow of L9.

Exercise 8.2

THE NOTEBOOKS, IN ORDER

- **00** — the timescales, the space–time sample, the reference
- **01** — the square, with a soft start
- **02** — the square, with the start built in
- **03** — the plate with its hole, in time
- **04** — the diffusivity recovered from sensors
- **05** — compares them, and the report

THE QUESTION EACH NOTEBOOK ANSWERS

- 01** where in time is the error?
- 02** what does a built-in start buy?
- 03** has the plate settled yet?
- 04** which sensors identify α ?

REPORT QUESTION, TO CONCLUDE

One loop produced the soft start, the hard start, the plate and the recovered diffusivity. What did time add to the loss, and which of your results would convince someone that the network is right?

Key Takeaways

TIME

what it is heat stored: one new term and input

function the field over a block of space–time

train, tune t_{end} from τ ; the start built in

vs steady an initial field, and no arrow of time

used for a device after a pulse, a cold start

PLATE AND INVERSE

what it is the plate of L8.1, started cold

function $t \neq N$: two conditions in one factor

train, tune both errors; sensors early

vs forward α trainable, found in the same run

used for properties you cannot measure

Next — L9.1: the same loop for a flow, where the nonlinearity is the difficulty.

Questions

1 – 2 · TIME

1 What does the heat equation add to steady conduction, and how does a PINN treat time?

A storage term, $\rho c_p \partial T / \partial t$, and so one initial field. The diffusivity $\alpha = k / \rho c_p$ says how fast a change spreads and sets the time constant τ . For a PINN time is one more input: the whole block of space and time is trained at once, and the solution can be read at any instant. *Liu ch. 4.1 – 4.3; Ex_08.2 notebooks 00 – 01*

2 What is causality, and how is it put back into training?

A solution at one instant follows from the earlier ones, but the loss treats all instants alike, so the network can fit late times first and carry an error at the start. First build the initial field into the trial solution; then causal weights, a split window, or more early points. *Liu ch. 4.2 – 4.4; Ex_08.2 notebooks 01 – 02*

3 – 4 · PLATE AND INVERSE

3 How is the transient plate set up, and how is its solution checked?

The trial solution $T(x, t)$ holds the initial field and the hole wall at once; the insulated edges stay a loss term. Report the relative and the absolute error, follow the hottest point in time, and use the energy balance once the field is steady. *Liu ch. 4.6; Ex_08.2 notebook 03*

4 How does an inverse problem differ from a forward one?

A parameter of the equation, here the diffusivity α , is trained beside the weights so that the solution fits measured data. It is optimised as $\log \alpha$, so it stays positive. Only the early part of the cooling curve constrains it: once the field has decayed, the data says nothing. *Liu ch. 4.6; Ex_08.2 notebook 04*

NEXT

L9.1 — Laminar Flow

The same loop again, on a velocity that carries itself: a nonlinear residual, and a pressure with no equation of its own.

BEFORE THE NEXT SESSION

read	Liu ch. 4.1 – 4.6
run	Ex_08.2 notebooks 00 to 04
bring	your error at $t = 0$, soft and hard
think about	your own problem's time constant

Block eight is finished: the plate, steady and in time, forward and inverse.